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merge sort

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Here are 5 Multiple Choice Questions (MCQs) on Merge Sort with their answers and explanations:

Merge Sort MCQs

1. Which sorting paradigm does Merge Sort primarily follow? a) Greedy Approach b) Dynamic Programming c) Divide and Conquer d) Backtracking

Answer: c) Divide and Conquer Explanation: Merge Sort recursively divides the array into two halves until individual elements are left (the 'divide' step), then merges those sorted halves back together (the 'conquer' step).

2. What is the time complexity of Merge Sort in its best, average, and worst cases? a) $O(n^2)$ b) $O(n \log n)$ c) $O(n)$ d) $O(\log n)$

Answer: b) $O(n \log n)$ Explanation: Merge Sort has a consistent time complexity of $O(n \log n)$ across all cases (best, average, worst) because it always divides the array into two halves and performs a linear-time merge operation.

3. Which of the following describes the auxiliary space complexity of Merge Sort? a) $O(1)$ b) $O(\log n)$ c) $O(n)$ d) $O(n \log n)$

Answer: c) $O(n)$ **Explanation:** Merge Sort requires an auxiliary array (or temporary space) of size $O(n)$ during the merging process to hold the sorted subarrays.

4. The core operation in Merge Sort, after recursively sorting the two halves, is to: a) Swap elements based on a pivot b) Partition the array around a chosen element c) Merge the two sorted subarrays into a single sorted array d) Insert an element into its correct position in a sorted part of the array

Answer: c) Merge the two sorted subarrays into a single sorted array **Explanation:** The "merge" step is where the two already sorted halves are combined to form a single larger sorted array. Options a) and b) relate to Quick Sort, and option d) relates to Insertion Sort.

5. Which statement accurately describes Merge Sort's properties? a) It is an in-place sort and unstable. b) It is an in-place sort and stable. c) It is not an in-place sort and unstable. d) It is not an in-place sort and stable.

Answer: d) It is not an in-place sort and stable. **Explanation:** Merge Sort is generally *not* an in-place sort because it requires $O(n)$ auxiliary space for the merging process. It is a *stable* sort, meaning it preserves the relative order of equal elements.